

CELL AND MOLECULAR BIOLOGY LABORATORY

From Gene Mutation to Disease

Investigating How DNA Sequence Changes Affect Protein Products and Human Phenotype

Name: Kris Bernadette S. Melon	Instructors: Mr. Abner A. Bucol Mrs. Lilibeth A. Bucol
Subject: BIO300 (Cell and Molecular Biology)	Section: B

Part I. Disease Background

Spinal and Bulbar Muscular Atrophy (SBMA) is an inherited neuromuscular disorder characterized by progressive weakness and wasting of the bulbar, facial, and limb muscles (La Spada, 2026). This disease is also known as Kennedy disease and primarily affects males because the responsible androgen receptor (AR) gene is located on the X chromosome. The SBMA results from the degradation and loss of lower motor neurons in the spinal cord and brainstem (Tanaka et al., 2012).

The genetic cause of SBMA is due to the increase in CAG trinucleotide repeats in exon 1 of the AR gene, which is located in the X chromosome at Xq12 (La Spada et al., 1992). The AR gene provides instructions for making the androgen receptor, a protein that responds to male sex hormones. The expanded CAG repeats produce an unusually long polyglutamine region in the androgen receptor, causing the X-linked neuromotor disorder spinal and bulbar muscular atrophy (Belikov et al., 2015).

Part II. Research the Disease

A. Disease

What is the disease or phenotype?	Spinal and Bulbar Muscular Atrophy, also known as Kennedy disease. It is a disorder that affects motor neurons.
What are its major clinical characteristics?	According to Martin (2025), spinal and bulbar muscular atrophy mainly affects males and usually starts in adulthood. SBMA causes progressive muscle weakness and muscle wasting, along with muscle twitching. People with SBMA may have difficulty walking, speaking, and swallowing. It can also cause weakness in the arms and legs, and some males may develop breast enlargement and reduced fertility.

Which cells, tissues, or organs are mainly affected?	SBMA mainly affects motor neurons, which are nerve cells that control muscle movement. According to Tanaka (2012), these neurons are found in the spinal cord and brainstem, causing weakness and wasting of facial, throat, and limb muscles.
What is its genetic basis?	SBMA is caused by an abnormal increase in CAG repeats in the androgen receptor (AR) gene on the X chromosome. The abnormal AR protein builds up inside the cell nucleus and contributes to motor neuron loss.
What is its inheritance pattern, if applicable?	SBMA has an X-linked inheritance pattern and mainly affects males. It can be passed through female carriers, who may have no symptoms or only mild symptoms.

B. Gene and Normal Protein

What is the official gene symbol?	AR
On which human chromosome is the gene located?	X Chromosome (Xq12)
What does the gene normally encode?	Androgen receptor (AR) protein
What is the normal biological function of the protein?	It receives signals from the male sex hormones and helps control certain genes.
Where in the cell is the protein normally found?	In the cytoplasm and nucleus.
In what biological pathway or cellular process does it participate?	It is involved in hormone signaling and gene regulation.

C. Documented Mutation

Gene	AR
Reference transcript	NM_000044.6
Exact variant notation	NM_000044.6:c.2296G>A (p.Ala766Thr)
Nucleotide change	c.2296G>A
Predicted protein change	NP_000035.2:p.Ala766Thr
Mutation type	Missense mutation

ClinVar accession	VCV000458363.13
Clinical interpretation	Pathogenic
Scientific reference	Batch, J. A., et al. (1992). Androgen receptor gene mutations identified by SSCP in fourteen subjects with androgen insensitivity syndrome. Human Molecular Genetics, 1(7), 497–503. PMID: 1307250. The study identified mutations in the androgen receptor gene in patients with complete and partial androgen insensitivity syndrome, showing that many of the mutations caused amino-acid changes in the receptor's ligand-binding domain.

Hypothesis

If a mutation is introduced, the normal AR coding sequence may alter the amino acid sequence and possibly affect the function of the androgen receptor. The documented c.2296G>A is predicted to replace alanine with threonine while keeping the reading frame the same, whereas the artificial GAA→GCA change is predicted to replace glutamic acid with alanine without causing a frameshift. These changes may modify the activity of the androgen receptor and may affect normal androgen-related processes.

Methods

This study examined the effects of different AR gene mutations using Galaxy by doing a comparison of the wild-type sequence with both a documented variant and an artificial mutation. The wild-type AR coding sequence (NM_000044.6) was obtained and imported into Galaxy as the reference sequence. This sequence was then translated using the Transeq tool to obtain the predicted androgen receptor protein. The WT sequence and translated protein were then analyzed based on their sequence length, reading frame, and amino-acid sequence.

For the documented variant, an AR c.2296G>A mutation was introduced by changing the G nucleotide into A at the c.2296 position in a copy of the WT coding sequence. This modified sequence was translated with Transeq and compared with the WT result to determine the amino-acid substitution and whether the mutation affected the reading frame or protein size. The translated sequence was also checked for any premature stop codon was produced.

An artificial mutation was then created by changing one nucleotide in the wild-type AR coding sequence, specifically changing GAA to GCA at coding sequence position 5. After translation, the sequence was compared with both the wild-type and documented mutation. The comparison focused on the type of mutation, changes in amino acids, reading frame, protein length, and occurrence of premature stop codons. These observations were

then used to assess how the different mutations could alter the AR protein and potentially influence its function.

Part III. Obtain the Normal Reference Sequence

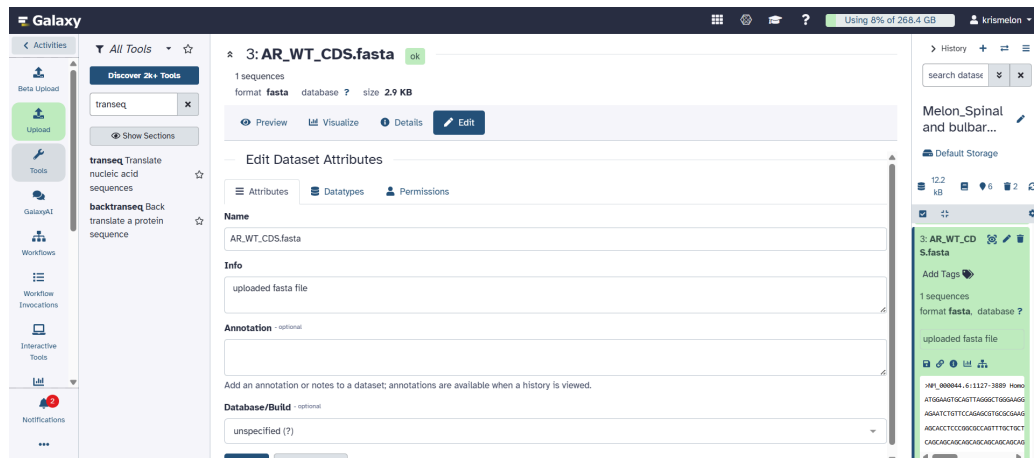


Figure 1. Uploading and renaming the wild-type AR coding sequence as AT_WT_CDS.fasta in Galaxy before translating it into the protein sequence.

Part IV. Import the Normal Sequence into Galaxy

The wild-type AR coding sequence (CDS) was imported into Galaxy and designated as the reference sequence for the analysis. A separate Galaxy history was created to organize the AR sequences, datasets, and analysis results, while the original WT sequence was preserved without modifications and used as the baseline for comparison with the documented and artificial mutations.

Part V. Establish the Wild-Type Control

CDS length	2763 bp
Predicted protein length	920 amino acids
Start codon	ATG
Stop codon	TGA
Reading frame	Frame 1
First 10 amino acids	MEVQLGLGRV
Last 10 amino acids	KVKPIYFHTQ

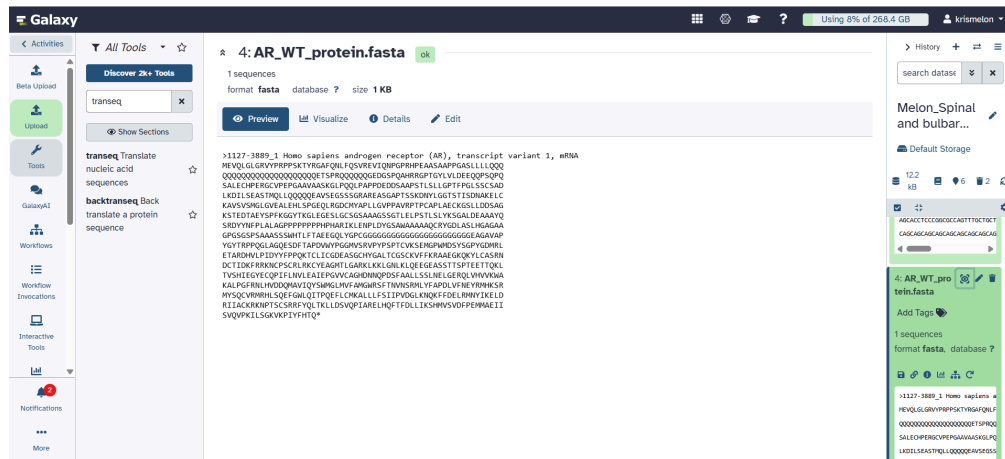


Figure 2. The wild-type AR coding sequence was translated into the predicted androgen receptor protein using the Transeq tool in Galaxy.

Part VI. Formulate a Mutation Hypothesis

Mutation and exact nucleotide change	NM_000044.6(AR):c.2296G>A (p.Ala766Thr)
Number of nucleotide(s) affected	1 nucleotide
Predicted mutation type	Substitution/Missense mutation
Predicted effect on the reading frame	No change in reading frame
Predicted effect on protein length	No change; remains 920 amino acids
Predicted effect on protein function	It may change the androgen receptor function due to the Ala766Thr substitution.

Part VII. Create the Mutant Sequence

Original nucleotide positions	c.2296
Original sequence	G
Mutant sequence	A
Number bases/inserted/deleted/substituted	of 1 nucleotide substituted
Mutation type	Missense

Part VIII. Translate the Mutant Sequence

Mutant CDS length	2,763 bp
Mutant protein length	920 amino acids
Reading frame	Frame 1

Location of the first amino-acid difference	Amino acid 766
Presence or absence of a premature stop codon	Absent
Approximate number of amino acids affected	1 amino acid
Mutant protein FASTA file name	AR_MT_CDS_protein.fasta

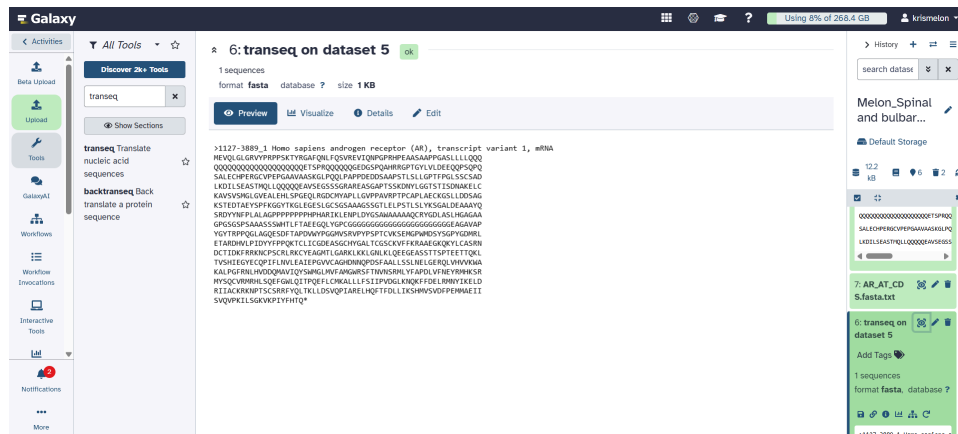


Figure 3. Galaxy Transeq output showing the predicted AR protein after the documented c.2296G>A (p.Ala766Thr) mutation was introduced into the coding sequence.

Part IX. Compare WT and Mutant Proteins

At what amino-acid position do the sequences first differ?	Position 766
Is only one amino acid affected?	Yes
Are multiple downstream amino acids changed?	No
Was an amino acid deleted or inserted?	No
Was a premature stop codon produced?	No
Did the reading frame change?	No
Did the protein length change?	No. The length remained the same.
Is the mutation missense, nonsense, frameshift, in-frame deletion/insertion, repeat expansion, or another type?	Missense

The c.2296G>A mutation substitutes G with A at nucleotide position 2296, changing the alanine at amino acid position 766 to threonine. Since the reading frame and overall protein length remain unchanged, the variant is classified as a missense mutation.

Part X. Explain the Molecular Consequence

The c.2296G>A mutation in the AR gene involves a substitution of a single nucleotide, from G to A. Since no bases were added or deleted, the reading frame did not change, and the CDS remained the same (2,763 bp). The mutation changed one amino acid at position 766, from alanine to threonine (p.Ala766Thr), making it a missense mutation. Alanine and threonine have different properties, so this change may affect how the androgen receptor works. Information from Clinvar shows that this mutation can affect the AR function. Since androgen receptors help respond to male hormones and control specific genes, a change in function can cause a defect in the androgen signaling pathway and lead to androgen resistance.

Part XI. Second Experiment: Create Your Own Mutation

Artificial Mutation	One-nucleotide substitution
Nucleotide Position	Position 5
Original Sequence	GAA
Mutant Sequence	GCA
Number of bases affected	1 nucleotide
Predicted Mutation Type	Missense
Predicted Effect in Reading Frame	No change; reading frame remains (Frame 1)
Predicted Effect on Protein	Glutamic acid at position 2 changes to alanine.

For the second experiment, I introduced an artificial mutation into the AR coding sequence by changing GAA to GCA at the coding sequence position 5. This single-nucleotide substitution is expected to change the second amino acid from glutamic acid to alanine while keeping the reading frame unchanged. The translated AR protein is therefore expected to retain the same overall length, with the amino acid sequence differing only at the affected position.

Part XII. Compare the Documented and Artificial Mutations

	WT Sequence	Documented mutation	Artificial mutation
CDS Length	2,763 bp	2,763 bp	2,763 bp
Protein Length	920 aa	920 aa	920 aa
Mutation Type	None	Missense	Missense
Reading frame	Frame 1	Unchanged	Unchanged

changed			
Premature stop codon appeared	No	No	No
Amino acids affected	None	Ala766 → Thr766	Glu2 → Ala2
Expected functional consequence	Normal androgen receptor function.	It may alter the androgen receptor function and androgen signaling.	It may alter receptor function because of the amino-acid substitution.

The wild-type AR sequence contains 2,763 bp and produces a 920-amino-acid protein. The documented mutation changes alanine 766 to threonine, while the artificial mutation changes glutamic acid 2 to alanine. Both are missense mutations that leave the reading frame and protein length unchanged but may have different effects on AR function.

Part XIII. Interpretation Questions

32. Why does the exact location of a mutation matter?

- The exact location of a mutation matters because different parts of a gene or protein have different functions. A mutation in an important region may have a greater effect on protein function.

33. Why can deleting three nucleotides produce a different result from deleting one or two nucleotides?

- Deleting three nucleotides removes one complete codon, so the reading frame stays unchanged. However, deleting one or two nucleotides shifts the reading frame and can change many amino acids.

34. Does every mutation change the amino-acid sequence? Explain.

- No. Some mutations are silent mutations, where the changed codon still codes for the same amino acid. Other mutations, such as missense and nonsense mutations, can change the amino-acid sequence.

35. Does every amino-acid substitution destroy protein function? Explain.

- No. Some amino-acid substitution destroys protein function. Some substitutions have little effect because the new amino acid may have similar properties or may occur in a region that is not critical for protein function.

36. Why can a frameshift affect many amino acids even if only one nucleotide was deleted?

- Deleting one nucleotide can shift the reading frame, causing the codons after the mutation to be read differently. As a result, many amino acids can change until a stop codon is reached.

37. Why might a premature stop codon produce a nonfunctional protein?

- A premature stop codon ends production too early. The resulting shortened protein may lack important regions needed for normal function.

38. Could a mutation affect protein function without greatly changing protein length?

- Yes, a mutation can affect protein function without changing its length. An example of this is missense mutation where it can replace one important amino acid while keeping the same number of amino acids.

39. Could a mutation cause disease without changing the protein sequence? Give a possible molecular mechanism.

- Yes, a mutation can cause diseases without changing the protein sequence. Such as a mutation in a regulatory region can change how much protein is produced.

40. What evidence from your analysis supports the proposed molecular mechanism of your disease?

- The analysis showed that the AR mutation changed one nucleotide and caused an Ala766→Thr amino-acid substitution. The reading frame and protein length remained unchanged, supporting the mechanism in which a single amino-acid change can alter AR function rather than simply shortening the protein.

41. Which conclusions are supported directly by your computational results, and which require evidence from published experimental studies?

- The computational results directly show the nucleotide change, amino-acid change, mutation type, reading frame, and protein length.

Part XV. GitHub Commit History

GitHub link:

<https://github.com/krismelon/Spinal-and-Bulbar-Muscular-Atrophy-AR-Mutation-Lab>

Galaxy link:

<https://usegalaxy.org/u/krismelon/h/melon-spinal-and-bulbar-muscular-atrophy-ar-mutation-lab>

Part XVI. Limitations

One limitation of this study was that the effects of the AR mutations were assessed mainly through computational analysis using Galaxy and not direct testing. Only one documented mutation, c.2296G>A, and one artificial mutation, GAA → GCA, were examined, so the findings may not apply to all AR mutations. Further laboratory experiments would be needed to determine how these mutations actually affect the androgen receptor's structure and function.

Part XVII. Conclusion

This study showed how changes in the AR gene can lead to changes in the androgen receptor protein. The documented mutation changed alanine at position 766 to threonine, while the artificial mutation change glutamic acid at position 2 to alanine. Both mutations were missense substitutions that did not alter the reading frame or protein length, showing that a single nucleotide change can result in an amino-acid substitution that may affect protein function.

Reference

AR gene: MedlinePlus Genetics. (n.d.). <https://medlineplus.gov/genetics/gene/ar/>

- Batch, J. A., Williams, D. M., Davies, H. R., Brown, B. D., Evans, B. A. J., Hughes, I. A., & Patterson, M. N. (1992). Androgen receptor gene mutations identified by SSCP in fourteen subjects with androgen insensitivity syndrome. *Human Molecular Genetics*, 1(7), 497–503. <https://doi.org/10.1093/hmg/1.7.497>
- Belikov, S., Bott, L. C., Fischbeck, K. H., & Wrangé, Ö. (2015). The polyglutamine-expanded androgen receptor has increased DNA binding and reduced transcriptional activity. *Biochemistry and Biophysics Reports*, 3, 134–139. <https://doi.org/10.1016/j.bbrep.2015.07.014>
- La Spada, A. R., Roling, D. B., Harding, A. E., Warner, C. L., Spiegel, R., Hausmanowa-Petrusewicz, I., Yee, W., & Fischbeck, K. H. (1992). Meiotic stability and genotype – phenotype correlation of the trinucleotide repeat in X-linked spinal and bulbar muscular atrophy. *Nature Genetics*, 2(4), 301–304. <https://doi.org/10.1038/ng1292-301>
- La Spada, A. (2026, January 8). *Spinal and bulbar muscular atrophy*. GeneReviews® - NCBI Bookshelf. <https://www.ncbi.nlm.nih.gov/books/NBK1333/>
- Martin, L. (2025, February 26). *What to know about spinal bulbar muscular atrophy (SBMA)*. <https://www.medicalnewstoday.com/articles/spinal-bulbar-muscular-atrophy>
- Tanaka, F., Katsuno, M., Banno, H., Suzuki, K., Adachi, H., & Sobue, G. (2012). Current status of treatment of spinal and bulbar muscular atrophy. *Neural Plasticity*, 2012, 1–8. <https://doi.org/10.1155/2012/369284>